



MTH101, Basic Linear Algebra

Monsoon 2025

Deadline: 21/11/2025, 23:59:59.

Homework Assignment-2

Throughout, V is an F -vector space.

- (1) Using results proved in previous lectures on injective linear transformations, explain why a homogeneous system of m equations and n unknowns have infinitely many solutions when $n > m$.
- (2) Let $T : V \rightarrow W$ be a one to one linear transformation of finite dimensional vector spaces. If $\dim_F(V) = \dim_F(W)$ then prove that T is an isomorphism.
- (3) Let $T : V \rightarrow W$ be an onto linear transformation of finite dimensional vector spaces. If $\dim_F(V) = \dim_F(W)$ then prove that T is an isomorphism.
- (4) Let $T : \mathbb{R}^n \rightarrow \mathbb{R}$ be a nonzero linear transformation of $\mathbb{R}$ -vector spaces, where $n \geq 2$. Find the dimension of $\ker(T)$.
- (5) For any two linear transformations $T_1 : V \rightarrow W$ and $T_2 : V \rightarrow W$ of F -vector spaces, define

$$T_1 + T_2 (v) = T_1(v) + T_2(v)$$

$$\alpha \star T_1 (v) = \alpha T_1(v).$$

Show $T_1 + T_2$ and $\alpha \star T_1$ are linear transformations from V to W and that the collection $\mathcal{L}(V, W)$ of all linear transformations from V to W forms an F -vector space under the above addition and scalar multiplication operations.

- (6) Find the matrix of the linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$, where $T(e_n) = e_1$ and for each $i = 1, 2, \dots, n-1$

$$T(e_i) = e_{i+1},$$

with respect to the basis $\{e_1, e_1 + e_2, e_1 + e_2 + e_3, \dots, e_1 + \dots + e_n\}$. Here $e_1, \dots, e_n$ are the standard unit vectors.

- (7) Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation such that $T(e_1) = e_1 + 4e_2$ and $T(e_2) = e_1 + e_2$. Find all possible eigenvalues of T and for each of the eigenvalue, find a corresponding eigenvector.